

Instructions to Project1

1. You are provided with the data: “drugs-combined.csv”, the files “best-so-far.ipynb”, and “feature-gen.py”
2. In “feature-gen.py”, topological features like: Persistent Entropy, Amplitude metrics (Wasserstein, Bottleneck, and Landscape), Betti curves were implemented.
3. Improve the accuracy of the Random Forest Classifier from the current value of approx 65% by adding more **topological/geometrical features** to “feature-gen.py” from the list below:
 - Spectral features: eigenvalues of weighted zero Laplacian and weighted 1-Laplacians, spectral range, algebraic connectivity.
 - Carlsson coordinates
 - Graph statistics: transitivity, cycle lengths, diameter, average path length, mean degree, max degree, etc.
4. Your total number of features for the Random Classifier must not exceed 120.
5. Any features used that **are not obtained from topology/geometrical data from graphs** will make your result invalid.
6. Any features added must be well understood as questions could be asked on them.